Advanced · Vector search infrastructure · Highlights · IBM Technology

Vector Databases: Storing Meaning Instead of Fields

Understand embeddings, distance metrics and the approximate-nearest-neighbour indexes that make semantic search fast.

For engineers who need to choose, tune or debug a vector store rather than just call one · 26 July 2026

The one- or two-page version: what matters, the topics it covers, and every term defined. Read the full lesson for the explanations and worked examples.

Watch the source video · Read the full lesson · Read the condensed version

What matters

- ✓ The semantic gap is the mismatch between structured storage and conceptual similarity — SQL predicates cannot express 'looks like this'.
- ✓ An embedding is an array of numbers positioned so that semantically similar items land close together and dissimilar ones land far apart.
- ✓ Individual dimensions in real embeddings are almost never interpretable; the geometry as a whole carries the meaning.
- ✓ Each modality has its own embedding models, and vectors from different models are never comparable.
- ✓ Embedding models extract progressively more abstract features layer by layer — edges to objects, words to meaning.
- ✓ Exhaustive search is O(N·d) and becomes impossible at scale, which is why vector indexes exist.
- ✓ HNSW navigates a multi-layer proximity graph; IVF partitions the space into clusters and searches only the nearest few.
- ✓ ANN indexes trade a small, measurable amount of recall for very large speed gains — and that recall must actually be measured.

What it covers

01 **The semantic gap** `0:00`

You can store an image in a relational database along with its metadata and tags, and still be unable to ask the questions that matter about it.

02 **Vectors as the representation** `2:11`

Represent each item as an array of numbers positioned so that similar items are near one another, and similarity search becomes arithmetic.

03 **What a vector actually encodes** `3:38`

Each position represents a learned feature. In a toy example those features are readable; in real systems they are not.

04 **Comparing two vectors** `4:20`

Two sunsets, one mountain and one beach: similar where they share meaning, different where they do not.

05 **Where embeddings come from** `5:48`

Embedding models trained on massive datasets, with each layer extracting progressively more abstract features.

06 **Similarity search, and why brute force runs out** `7:16`

Comparing a query against every vector is exact and simple, and becomes impossible somewhere in the millions.

07 **ANN indexing: HNSW and IVF** `8:00`

Two ways to avoid looking at everything — navigate a proximity graph, or search only the nearest clusters.

08 **Vector databases inside RAG** `8:41`

Store document chunks as embeddings, find the relevant ones by similarity, hand them to a language model.

Glossary

Semantic gap — The mismatch between how computers store data — files, fields, tags — and how humans understand its meaning.

Vector database — A store built to hold embeddings and retrieve them by similarity, with the indexing and filtering machinery that makes that fast.

Euclidean distance — Straight-line distance between two points in the vector space. Sensitive to magnitude as well as direction.

L2 normalisation — Scaling a vector to unit length, which makes cosine, dot product and Euclidean produce identical rankings.

HNSW — Hierarchical Navigable Small World: a multi-layer proximity graph navigated greedily from sparse upper layers down to the dense base.

ef_search — HNSW's query-time beam width. The main dial for trading recall against latency, changeable without rebuilding the index.

Quantisation — Compressing vectors to fewer bits per dimension to cut memory, at some cost in precision. Often paired with IVF at large scale.

Static vs contextual embedding — Static models give each word one fixed vector; contextual encoders produce representations that depend on the surrounding text.

Vector embedding — An array of numbers representing an item, positioned so that semantically similar items land close together.

Cosine similarity — The cosine of the angle between two vectors. Ignores magnitude, which is usually what you want for text.

Dot product — A similarity measure combining alignment and magnitude. Equivalent to cosine when vectors are normalised.

Approximate nearest neighbour (ANN) — Algorithms that find vectors very likely to be the closest, trading a little accuracy for a large speed gain.

IVF — Inverted File Index: partitions the space into clusters and searches only the nearest few, controlled by nprobe.

nprobe — How many IVF clusters a query searches. One is fastest and least accurate; all of them is equivalent to exhaustive search.

Recall@k — The fraction of true nearest neighbours an approximate index actually returns. The ceiling on any system built on it.